

EduVision_DV: Dashboard Architecture & UI Storyboard

Milestone 2 / Module 4: UI/UX Specification, KPI Wireframes & Navigation Architecture | **Author: Sujay S** (Infosys Springboard)

Project Statement: EduVision_DV is an integrated, executive-grade Higher Education Performance Intelligence System developed in Tableau Desktop. It synthesizes data across 1,503 world-class institutions and macroeconomic indicators to deliver four interconnected dashboards linked through standardized keys (university_id, country_id, year) and dynamic filter actions.

Tableau Workbook Dashboard Architecture Overview

Dashboard Name	Target Audience	Core Visual Components	Interlinking Actions & Data Passing
1. University Overview (Landing Page)	University Chancellors, Academic Deans, Prospective Students	6 Top KPI Metric Banners - Top 10 Global Ranking Bar Chart - Academic Reputation vs. Overall Score - World Choropleth Distribution Map - Regional Distribution Donut Chart	Clicking any university passes university_id to Dashboards 2 & 3. Clicking a nation passes country_id to Dashboard 4.
2. Research Analytics	Provosts, Research Deans, National Science Foundations	- Research Environment & Citation Banners - Research Productivity Index Ranking - Citations vs. Research Environment Scatter - International Research Collaboration Index	Pre-filtered by selected university_id with global and regional peer benchmark overlay lines.
3. Student Analytics	Admissions Directors, International Recruiters	- Student Headcount & Ratio Banners - International Student % Distribution - Students-per-Staff Capacity Benchmark - Female:Male Gender Ratio Breakdown	Pre-filtered by university_id; includes one-click transition to national context on Dashboard 4.
4. Country Comparison	Education Ministers, Macroeconomists, Policy Analysts	- National Higher Ed Capacity Bar Chart - Average National Ranking & Score - Education Spend (% GDP) vs Performance - Tertiary Enrollment Historical Trends	Receives country_id from preceding dashboards; allows multi-country benchmark selection.

Design System & UI Color Standards

Theme Element	Hex Code	Visual Role & Rationale
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Primary Background	#0F172A (Slate 900)	Deep navy background reduces ocular fatigue, elevates chart mark contrast, and delivers a sleek executive UI.
Card Containers	#1E293B (Slate 800)	Slightly elevated dark panels that segment charts into modular analytical cards.
Academic Excellence Accent	#10B981 (Emerald Green)	Highlights Academic Reputation, Global Rank, and overall institutional excellence.
Research Output Accent	#06B6D4 (Cyan)	Highlights Research Environment, Citation Impact scores, and Research Productivity.
International Diversity Accent	#F59E0B (Amber / Gold)	Highlights International Student percentages and cross-border student mobility.
Faculty Mentorship Accent	#8B5CF6 (Purple)	Highlights Faculty-to-Student ratios, staff headcounts, and teaching indicators.

Dashboard 1 Wireframe: University Overview (Landing Page)

High-level institutional scorecard and global performance benchmarking. Serves as the primary entry point for institutional drill-downs.

TOP HEADER BAR: EduVision DV Higher Education Analytics Filters: [Region: All ▼] [Country: All ▼] [University: All ▼] [Year: 2025 ▼]	
KPI 1: Global Rank #1 (MIT) Top Tier World	KPI 2: Overall Score 100.0 / 100 QS Composite Benchmark
KPI 3: Academic Reputation 100.0 / 100 Peer Survey Benchmark	KPI 4: Research Impact 100.0 / 100 Citations per Faculty
CHART A: Top 10 Global Universities Horizontal ranked bar chart showing Overall Score (MIT: 100, Imperial: 98.5, Oxford: 96.9, Harvard: 96.8, Cambridge: 96.7). Filter Action: Clicking any bar filters Dashboards 2 & 3 via university_id.	CHART B: Academic Reputation vs. Overall Score Scatter plot showing correlation between peer survey esteem and composite score. Highlights high-efficiency institutions. Filter Action: Selecting marks filters detail tables.
CHART C: Global University Distribution Map World Choropleth Map highlighting institutional concentration across 106 nations. Filter Action: Clicking a nation passes country_id to Dashboard 4.	CHART D: Regional Share of Top Universities Donut Chart displaying continental distribution (Europe 42%, North America 34%, Asia 18%, Oceania 4%, Latin America 2%).

Dashboard 2 Wireframe: Research Analytics & Scholarly Output

Deep-dive into research environment, citation performance, and cross-border collaborative research networks.

TOP HEADER BAR: EduVision DV Research Intelligence Suite Active Institution: [University of Oxford (Pre-Filtered)]	
Research Environment Score: 99.6 / 100 THE Research Infrastructure & Income	Citation Impact Score: 99.0 / 100 Normalized Peer Citation Velocity
Research Productivity Index (KPI 6): 95.44 / 100 Formula: 0.50*Res + 0.30*Cit + 0.20*Network	International Research Network: 97.8 / 100 Cross-Border Collaborative Co-Authorship
CHART A: Top Research Hubs by Productivity Index Ranked bar chart highlighting Oxford, Harvard, Stanford, Cambridge, MIT, and Caltech with global average benchmark lines.	CHART B: Citation Impact vs. Research Environment Scatter plot identifying high-efficiency research institutions converting moderate research funding into high citation influence.

Dashboard 3 Wireframe: Student Analytics & Diversity

Evaluates student demographics, actual faculty-to-student mentorship ratios, and campus international diversity.

TOP HEADER BAR: EduVision DV Student Demographics & Diversity Active Institution: [University of Oxford]	
Total FTE Students: 20,700 <i>Full-Time Equivalent Enrollment Headcount</i>	Faculty-to-Student Ratio (KPI 3): 1 : 10.6 <i>Actual Students per Academic Staff Member</i>
International Students % (KPI 4): 42.0% <i>Empirical Cross-Border Student Representation</i>	Female : Male Ratio: 48 : 52 <i>Campus Gender Demographic Breakdown</i>
CHART A: International Student % Distribution Histogram / ranked bar showing cross-border student share. Identifies highly globalized universities (Imperial: 61%, Oxford: 42%).	CHART B: Students-per-Staff Capacity Benchmark Bar chart comparing faculty mentorship capacity against global mean (10.6 vs global mean of 18.7).
CHART C: Gender Parity Across Disciplines Stacked bar chart analyzing female vs male student enrollment across peer institutions.	CHART D: Enrollment Scale vs. International Share Bubble plot sizing circles by total FTE student body and positioning by international percentage.

Dashboard 4 Wireframe: Country Comparison (World Bank EdStats Benchmarking)

Macroeconomic education intelligence linking national public spending and tertiary enrollment with top-tier university capacity.

TOP HEADER BAR: EduVision DV Macro Education Policy & Country Benchmark Selected Country: [United Kingdom]	
National University Capacity: 90 Universities <i>Ranked World-Class Institutions</i>	Average National Score: 48.6 / 100 <i>Mean QS Overall Score for Nation</i>
Public Education Spend: 5.2% of GDP <i>World Bank Government Expenditure Benchmark</i>	Gross Tertiary Enrollment: 66.8% <i>Total Tertiary Participation Rate</i>
CHART A: World-Class University Capacity by Nation Ranked bar chart of top nations by number of ranked universities (USA: 197, UK: 90, China: 71, Germany: 49, India: 46, Japan: 49).	CHART B: Public Education Spend (% GDP) vs. Institutional Performance Scatter plot evaluating national return on investment: public spending percentage plotted against average university score.
CHART C: Tertiary Enrollment Trajectories (2010–2023) Multi-line historical trend comparison of gross tertiary enrollment rates across selected peer nations.	CHART D: National Adult Literacy & Secondary Completion Comparative bar chart displaying baseline foundational education indicators.

Interlinking Architecture & End-to-End User Journey

Demonstrates how user interaction flows dynamically across all four dashboards through Tableau filter actions.

Step #	User Action & Dashboard	Tableau Mechanism Triggered	Visual Outcome & Analytical Value
Step 1	User opens Dashboard 1 (University Overview) and clicks University of Oxford row on the Top 10 bar chart.	Tableau Filter Action triggered on mark selection, passing university_id = 'U0003' to target sheets.	Top KPI cards update immediately to Oxford's profile (Rank: 3, Overall Score: 96.9, Academic Reputation: 100.0).
Step 2	User clicks ' Research Analytics →' in the persistent top navigation bar.	Navigation Button switches active dashboard view to Dashboard 2 while maintaining university_id filter.	Dashboard 2 opens pre-filtered for Oxford, revealing Research Productivity Index of 95.44 , Citations of 84.8, and Collaboration Network of 97.8.
Step 3	User clicks ' Student Analytics →' button to examine campus mentorship and student demographics.	Navigation Button switches view to Dashboard 3 with persistent university_id context.	Shows Oxford's student profile: actual student-to-staff ratio of 10.6:1 , enrollment of 20,700, and 42.0% international students.
Step 4	User clicks ' United Kingdom ' or navigates to Dashboard 4 (Country Comparison) .	Tableau Country Filter Action passes country_id = 'C0002' to Macro Education fact tables.	Dashboard 4 displays the UK's 90 ranked universities, national tertiary enrollment trends (2010–2023), and 5.2% GDP public education spend.

Tableau Filter Action & Parameter Configuration Summary

Filter Action Name	Source Sheet / Dashboard	Target Sheets / Dashboard	Passed Fields & Behavior
Filter_University_Drilldown	Top 10 Global Ranking Bar Chart ('Dashboard 1')	All views on `Dashboard 2` & `Dashboard 3`	Passes university_id. Clearing selection restores all institutional records.
Filter_Country_Drilldown	Global University Distribution Map ('Dashboard 1')	All views on `Dashboard 4 (Country Comparison)`	Passes country_id. Displays national university density and macro spend stats.
Nav_Button_Actions	Top Navigation Bar on all 4 Dashboards	Target Dashboard (1, 2, 3, or 4)	Seamless one-click tab transitions without resetting global filter parameters.